

# NORDSTROM

## 2025 Retail Sales Analysis | Executive Summary

**Tools Used:** SQL (data extraction & transformation) · Tableau (visualization)

|                                                      |                                                      |                                                                |
|------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| <b>37.11%</b><br>of total profit from top 5 products | <b>\$63.7K</b><br>profit generated by Leather Loafer | <b>~47%</b><br>profit margin, consistent across discount tiers |
|------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|

### OBJECTIVE

This project pulls together transaction, product, customer, discount, and inventory data to answer specific questions about how the business performed in 2025 — where sales and profit came from, how customers behaved, which products and discounts pulled their weight, and how inventory moved. The goal was to figure out what's actually driving performance, flag the risks, and turn that into recommendations management can act on.

### BUSINESS QUESTIONS

The project was developed around a defined set of business questions covering profitability, customer behavior, sales consistency, discount effectiveness, inventory management, and performance trends. These questions guided the SQL analysis and subsequent Tableau dashboard development.

### DATA & METHODOLOGY

Transaction-level retail data was analyzed in SQL, covering everything from initial data exploration to segmentation and profitability calculations. Departmental sales stability was measured using the coefficient of variation (CV) to distinguish consistent performers from volatile ones. Findings were then built out into interactive Tableau dashboards for both executive and operational use.

### KEY FINDINGS

- Women's generated the highest total sales and profit, while Kids delivered the strongest profit margin.
- The top five products alone accounted for 37.11% of total profit, pointing to meaningful profit concentration in a small part of the catalog.
- Men's had the most stable monthly sales (~5% CV), compared with Women's (~6%) and Kids (~10%), which fluctuated more month to month.
- Profit margins held steady near 47% across all discount groups — suggesting discounting isn't clearly moving the needle on profitability and warrants a closer, transaction-level look.

### RECOMMENDATIONS

- Prioritize inventory availability for the company's highest-profit products, starting with the Leather Loafer (~\$63.7K in profit) and the rest of the top-profit group — since the top five products account for 37.11% of total profit, a stockout on any of them puts real profit at risk.
- Investigate the Kids department's higher sales variability, whether it's seasonality, product mix, promotions, or inventory driving the swings.

- Reevaluate discount strategy at the product and transaction level — margins held steady near 47% regardless of discount level, with no clear sales lift from heavier discounting.
- Weigh profitability and inventory movement alongside sales, not sales alone, to flag real replenishment priorities versus overstock risk.

## BUSINESS QUESTIONS

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To provide a structured approach to the analysis, the project was organized around the following business questions. We investigated each question using SQL and, where appropriate, translated the results into Tableau visualizations to support management decision-making.

**1. Which departments generate the highest sales and profit, and which have the strongest profit margin**

**Key findings:** The women's department has the strongest overall sales and profit, with annual sales of \$758,423.21 and profits of \$357,187.87. While women had the strongest sales, the Kids department had the highest profit margin at 47.2%, compared with 47.1% for women's and 46.6% for men's.

**2. Which department generates the most consistent sales, and which department has the greatest volatility?**

**Key findings:** The men's department, on the other hand, had the most consistent sales with the lowest coefficient of variation of 5.24%. Women followed at 5.86%, and kids had the greatest volatility of 9.97%.

**3. Which product and product category contribute most to overall revenue and profitability?**

**Key findings:** The top 5 products generated around 37.11% of total company profits, showing that profitability was concentrated among a small number of products; Leather loafers, of which, generated around \$63,700 in profit in one department, making it the strongest individual profit contributor.

**4. How do customer segments differ in purchasing behavior, spending, and profitability?**

**Key findings:** Loyalty customers were the largest segment, with 6,231 transactions, \$649,309.51 in sales, and \$304,814.08 in profit, with an average transaction value of \$104.21. Returning customers generated 4,545 transactions, \$461,177.22 in sales, and \$218,173.39 in profit, averaging \$102.13 per transaction. New customers generated 4,224 transactions, \$436,679.43 in sales, and \$205,069.83 in profit, averaging \$103.38 per transaction. Profit margins were extremely similar across all three groups at approximately 46.94%–47.00%.

**5. How do discounts affect sales volume, revenue, and profit margins?**

**Key findings:** Profit margins remained around 47% across the different discount groups, with no meaningful improvement in profitability as discount levels increased. This indicates that heavier discounts did not produce a clear margin advantage and should be evaluated based on their effect on sales volume and revenue rather than assumed to improve profitability.

**6. What actionable opportunities can management identify to improve revenue profitability and inventory efficiency?**

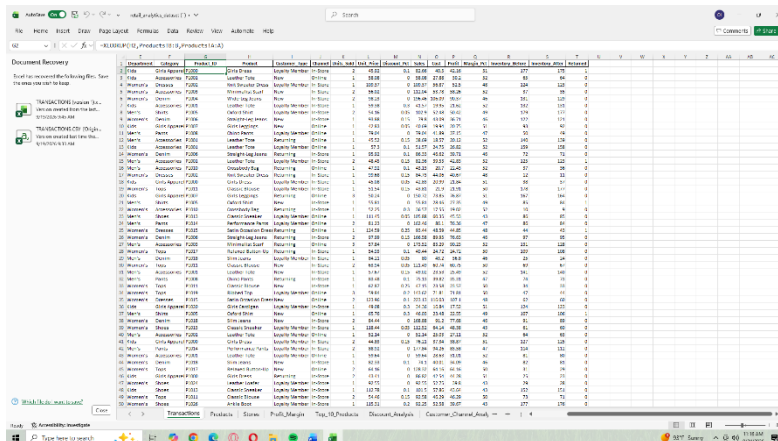
**Key findings:** Prioritize inventory availability for top-profit products, starting with the Leather Loafer and the rest of the top five, since they account for 37.11% of total profit and a stockout on any of them puts meaningful profit at risk. Investigate the drivers behind the Kids department's higher sales volatility (~9.97% CV) — whether seasonality, product mix, promotions, or inventory timing — to smooth out swings relative to Men's and Women's. Reevaluate discount strategy at the product and transaction level, since margins held steady near 47% regardless of discount depth, showing no clear profitability lift from heavier discounting. Continue building out customer segment strategies, since Loyalty, Returning, and New customers show similar margins but differing transaction volumes and values, suggesting acquisition and retention efforts could be tailored by segment. Complete the inventory risk and seasonality analyses to round out a full-year view before finalizing efficiency recommendations.

## SQL analysis and methodology

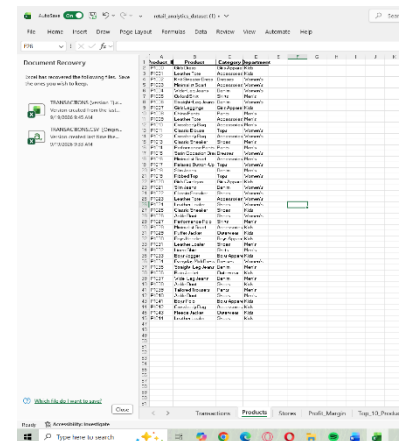
Microsoft Excel was used during the initial data preparation stage to review and clean the raw datasets, identify inconsistencies, and create PivotTables for preliminary analysis and validation. The cleaned and structured data was then imported into SQL, where transaction, product, customer, discount, and inventory data were combined and analyzed to calculate key performance metrics and evaluate profitability, customer behavior, sales consistency, discount effectiveness, and inventory performance. The final analytical results were then visualized in Tableau through interactive dashboards designed to support management decision-making.

### 1. Data Sets (Excel)

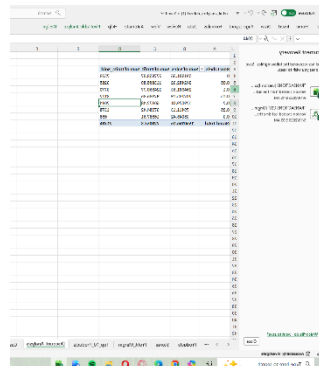
Two Excel datasets were used for the analysis. The **Transactions** dataset contained every purchase recorded across all stores throughout 2025, while the **Products** dataset contained product-level information for the merchandise sold. We cleaned and organized the datasets in Excel, using XLOOKUP to match product information to transaction records. We then created PivotTables to validate totals, explore initial trends, and prepare the data for deeper analysis in SQL.



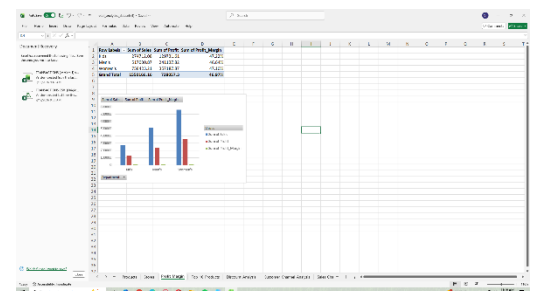
Transaction Data Set with X-lookup



Product Data Set



Pivot Table



Pivot Table

### 2. Data Preparation Workflow-

1. Validate- Looking through all the data using data validation techniques to make sure no duplicates, missing, or null values are in the dataset.
2. Join- Combined the first dataset with the second dataset to get a 3NF finished dataset.
3. Transform- Used pivot tables to create visualizations of the data.
4. Aggregate- Summarized all data and cross-checked before creating SQL queries.

### 3. DATA CALCULATIONS / FORMULAS USED

Metric – Sum(sales)

Total Sales- Sum(profit)

Total Profit- Total profit / Total Sales

Profit Margin- Total Sales / Transactions

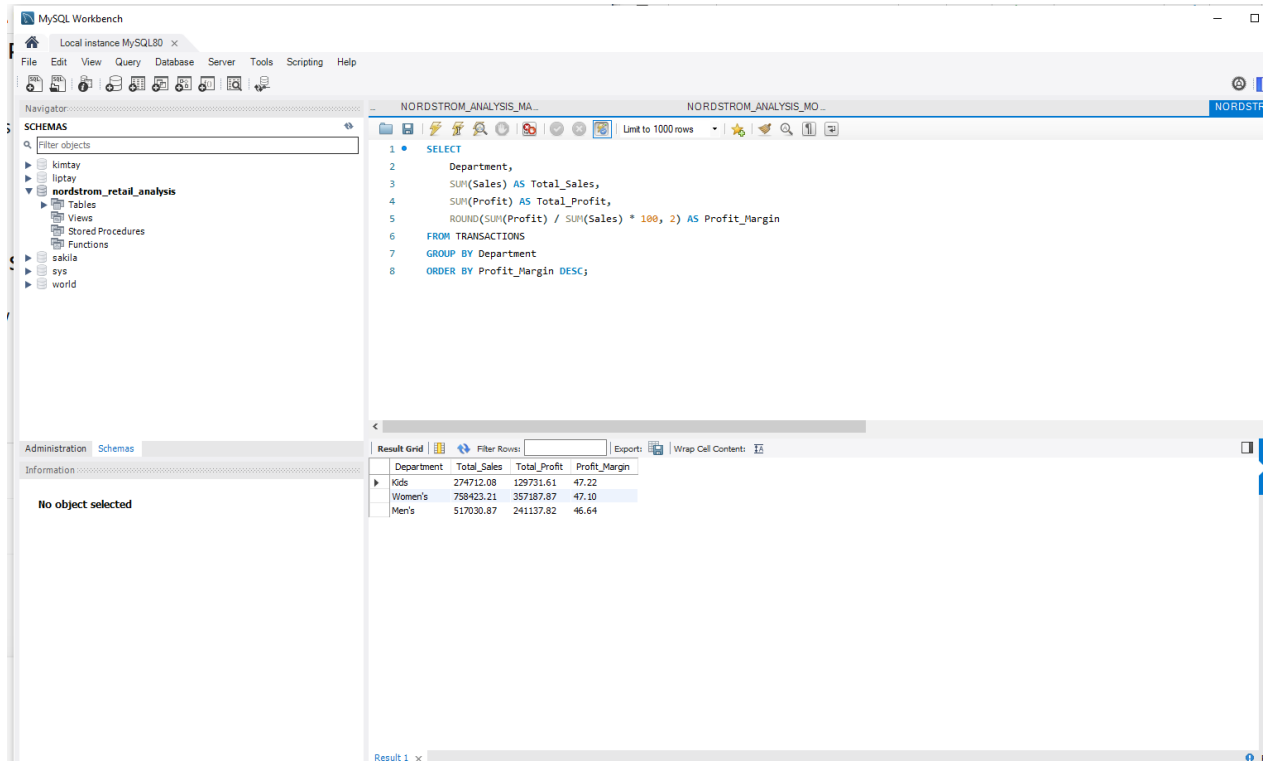
Avg Transaction Value- Total Sales / Transactions

Coefficient of variation- STDDEV (Monthly Sales) / Avg (Monthly Sales)

## SQL ANALYSIS

Department Performance-

Question: Which departments generate the highest sales and profit, and which have the strongest margins?



The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays a tree view with 'nordstrom\_retail\_analysis' expanded. The main editor shows a SQL query:

```
1 SELECT
2   Department,
3   SUM(Sales) AS Total_Sales,
4   SUM(Profit) AS Total_Profit,
5   ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS Profit_Margin
6 FROM TRANSACTIONS
7 GROUP BY Department
8 ORDER BY Profit_Margin DESC;
```

Below the query, the 'Result Grid' shows the following data:

| Department | Total_Sales | Total_Profit | Profit_Margin |
|------------|-------------|--------------|---------------|
| Kids       | 274712.08   | 129731.61    | 47.22         |
| Women's    | 758423.21   | 357187.87    | 47.10         |
| Men's      | 517030.87   | 241137.82    | 46.64         |

Answer: Women had the highest sales with \$758,423.21 in sales, but the Kids department had the best profit margin with 47.22%

Sales Consistency & Volatility

Question: Which department generates the most consistent monthly sales?

SQL Query Editor: NORDSTROM\_ANALYSIS\_MA\_

```
1 SELECT
2     Department,
3     ROUND(AVG(Monthly_Sales), 2) AS Avg_Monthly_Sales,
4     ROUND(STDDEV(Monthly_Sales), 2) AS Sales_Std_Dev
5 FROM (
6     SELECT
7         Department,
8         MONTH(Order_Date) AS Month_Number,
9         SUM(Sales) AS Monthly_Sales
10    FROM TRANSACTIONS
11   GROUP BY
12        Department,
13        MONTH(Order_Date)
14 ) AS Monthly_Department_Sales
15 GROUP BY Department
16 ORDER BY Sales_Std_Dev ASC;
```

Result Grid

| Department | Avg_Monthly_Sales | Sales_Std_Dev |
|------------|-------------------|---------------|
| Kids       | 22892.67          | 2240.28       |
| Men's      | 43085.91          | 2255.98       |
| Women's    | 63201.93          | 3703.23       |

Result 1 x

Output

| # | Time     | Action | Message                                                                                                                                                    |
|---|----------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 13:09:13 | SELECT | Department, SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit, ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS Coefficient_of_Variation 3 row(s) returned |
| 2 | 13:14:00 | SELECT | Department, ROUND(AVG(Monthly_Sales), 2) AS Avg_Monthly_Sales, ROUND(STDDEV(Monthly_Sales), 2) AS Sales_Std_Dev 3 row(s) returned                          |

SQL Query Editor: NORDSTROM\_ANALYSIS\_MO\_

```
1 SELECT
2     Department,
3     ROUND(AVG(Monthly_Sales), 2) AS Avg_Monthly_Sales,
4     ROUND(STDDEV(Monthly_Sales), 2) AS Sales_Std_Dev,
5     ROUND(
6         STDDEV(Monthly_Sales) / AVG(Monthly_Sales) * 100,
7         2
8     ) AS Coefficient_of_Variation
9 FROM (
10    SELECT
11        Department,
12        MONTH(Order_Date) AS Month_Number,
13        SUM(Sales) AS Monthly_Sales
14    FROM TRANSACTIONS
15   GROUP BY
16        Department,
17        MONTH(Order_Date)
18 ) AS Monthly_Department_Sales
19 GROUP BY Department
20 ORDER BY Coefficient_of_Variation ASC;
```

Result Grid

| Department | Avg_Monthly_Sales | Sales_Std_Dev | Coefficient_of_Variation |
|------------|-------------------|---------------|--------------------------|
| Men's      | 43085.91          | 2255.98       | 5.24                     |
| Women's    | 63201.93          | 3703.23       | 5.86                     |
| Kids       | 22892.67          | 2240.28       | 9.79                     |

Answer: Men produced the most stable monthly sales. Kids' CV of 9.97% indicates large variation in sales on a month-to-month basis.

### Product Profitability-

Question: Which products generated the most profit for the company?

Local Workbench

Local instance MySQL80 x

File View Query Database Server Tools Scripting Help

objects

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object selected

NORDSTROM\_ANALYSIS\_MO ...

Limit to 1000 rows

```
1 • SELECT
2     Department,
3     Product,
4     Total_Profit,
5     Profit_Rank
6 FROM (
7     SELECT
8         Department,
9         Product,
10        SUM(Profit) AS Total_Profit,
11        RANK() OVER (
12            PARTITION BY Department
13            ORDER BY SUM(Profit) DESC
14        ) AS Profit_Rank
15 FROM TRANSACTIONS
16 GROUP BY Department, Product
17 ) AS Ranked_Products
18 WHERE Profit_Rank <= 3
19 ORDER BY Department, Profit_Rank;
```

Result Grid

|   | Department | Product              | Total_Profit | Profit_Rank |
|---|------------|----------------------|--------------|-------------|
| ▶ | Kids       | Ankle Boot           | 13404.29     | 1           |
|   | Kids       | Leather Loafer       | 12349.74     | 2           |
|   | Kids       | Rain Jacket          | 12018.90     | 3           |
|   | Men's      | Leather Loafer       | 22492.32     | 1           |
|   | Men's      | Ankle Boot           | 20766.43     | 2           |
|   | Men's      | Classic Sneaker      | 19946.54     | 3           |
|   | Women's    | Satin Occasion Dress | 31606.63     | 1           |
|   | Women's    | Knit Sweater Dress   | 31071.24     | 2           |
|   | Women's    | Everyday Midi Dress  | 31019.25     | 3           |

Answer - The top five products generated approximately 37.11% of total company profit, indicating that profitability was concentrated among a relatively small group of products. The Leather Loafer was the strongest individual profit contributor, generating approximately \$63,700 in profit.

Question: Which Department had the most consistent sales, and which department had the least consistent?

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays a tree view of databases: kimtay, liptay, nordstrom\_retail\_analysis (selected), sakila, sys, and world. The 'nordstrom\_retail\_analysis' database is expanded, showing Tables, Views, Stored Procedures, and Functions. The main editor displays a SQL query (lines 1-20) that calculates the coefficient of variation for each department. The query uses the 'TRANSACTIONS' table, grouping by department and month, and calculating the average and standard deviation of monthly sales. The results are ordered by the coefficient of variation in ascending order. The 'Result Grid' at the bottom shows the output of the query, with columns for Department, Avg\_Monthly\_Sales, Sales\_Std\_Dev, and Coefficient\_of\_Variation. The results are as follows:

| Department | Avg_Monthly_Sales | Sales_Std_Dev | Coefficient_of_Variation |
|------------|-------------------|---------------|--------------------------|
| Men's      | 43085.91          | 2255.98       | 5.24                     |
| Women's    | 63201.93          | 3703.23       | 5.86                     |
| Kids       | 22892.67          | 2240.28       | 9.79                     |

Answer - The Men's department had the most consistent monthly sales, with the lowest coefficient of variation (CV) at 5.24%. The Kids department had the least consistent sales, with the highest CV at 9.97%. Women's fall in between at 5.86%.